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Comorbidity-based framework for Alzheimer's disease classification using graph neural networks

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Alzheimer's disease (AD), the most prevalent form of dementia, requires early prediction for timely intervention. Current deep learning approaches, particularly those using traditional neural networks, face challenges such as handling high-dimensional data, interpreting complex relationships, and managing data bias. To address these limitations, we propose a framework utilizing graph neural networks (GNNs), which excel in modeling relationships within graph-structured data. Our study employs GNNs on data from the Alzheimer's Disease Neuroimaging Initiative for binary and multi-class classification across the three stages of AD: cognitively normal (CN), mild cognitive impairment (MCI), and Alzheimer's disease (AD). By incorporating comorbidity data derived from electronic health records, we achieved the most effective multi-classification results. Notably, the GNN model (Chebyshev Convolutional Neural Networks) demonstrated superior performance with a 0.98 accuracy in multi-class classification and 0.99, 0.93, and 0.94 in the AD/CN, AD/MCI, and CN/MCI binary tasks, respectively. The model's robustness was further validated using the Australian Imaging, Biomarker & Lifestyle dataset as an external validation set. This work contributes to the field by offering a robust, accurate, and cost-effective method for early AD prediction (CN vs. MCI), addressing key challenges in existing deep learning approaches.

Alzheimer's disease (AD) is a progressive neurological disorder marked by memory loss, cognitive decline, and behavioral changes, ultimately resulting in severe impairment in daily functioning¹. As per the World Health Organization (WHO), approximately 10 million new cases of dementia arise annually among the 55 million individuals worldwide affected, with AD constituting 60-70% of these cases². AD presents a multifaceted challenge characterized by its intricate progression, lacking clear demarcations between stages and often evading detection until advanced phases; it has three defined stages: preclinical (cognitive normal (CN))³, mild cognitive impairment (MCI)⁴, and dementia due to AD⁵. Early detection of the MCI stage is crucial in AD management, as it allows for interventions that can slow cognitive decline by identifying individuals at risk and implementing appropriate lifestyle changes, despite the absence of a cure⁶.

The classification of AD using machine learning (ML) and deep learning (DL) approaches has been the subject of a significant amount of research⁷⁻⁹, where data modalities such as neuroimaging^{10,11} have been widely used alone or together with other modalities¹²⁻¹⁴. While imaging modalities play a crucial role in confirming structural changes associated with AD, their use for early detection has drawbacks due to the high costs and accessibility issues. Alternatively, electronic health records (EHRs) provide comprehensive patient information, longitudinal assessment, accessibility, cost-effectiveness, and holistic care without specialized equipment or additional cost^{15,16}. EHRs have been successfully employed in predicting various diseases¹⁷⁻²⁰. Comorbidity data in the structured EHRs presents opportunities for data mining using machine learning algorithms to help develop more accurate and efficient methods for AD classification, potentially leading to earlier diagnosis and intervention. Comorbidity data refers to the documentation of multiple medical conditions or diseases that coexist in a patient's health record²¹. They have been linked to an increased risk of death in AD and dementia patients and can be used to provide a comprehensive and multidimensional approach to AD prediction²². Evidence from various studies has established associations between AD and other chronic diseases, such as diabetes²³, cardiovascular disease²⁴, and

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depression²⁵. Despite EHR data being utilized in AD survival analysis studies²⁶, its potential in AD classification has been largely overlooked in the existing literature. Thus, the inclusion of EHR-derived comorbidities is deemed essential due to their role in introducing complexity to AD classification, primarily due to shared risk factors with other comorbidities.

Recently, EHR data conversion into low-dimensional vectors has garnered attention in the field of healthcare, enabling machine learning algorithms to undertake predictive tasks²⁷. However, hidden relationships exist between chronic diseases, their comorbidities, and patients, potentially impacting the accuracy of predictive models. To address this issue, Social Network Analysis (SNA) has been applied to healthcare data where they utilized network features extracted from the patient network, combined with individual patient characteristics, in ML algorithms for predicting the risk of chronic diseases²⁸. Many of these studies use basic ML algorithms like logistic regression and random forests, and require redevelopment and feature recalculation when new data is introduced. In contrast, GNN-based approaches can seamlessly incorporate new data without the need for such recalculations, making GNNs a very attractive alternative for complex multi-classification problems on EHR data. GNNs offer a promising solution by setting up relationships between patients in a neighborhood graph, where nodes represent patients and edges represent similarities based on risk factors like age, gender, and genes²⁹. This approach has shown great potential in the medical field, particularly in integrating diverse information for disease diagnosis. In³⁰, a patient record graph was constructed that integrates medical knowledge base and electronic medical records. The framework was complemented by a neural graph encoder, which generated node embedding for the constructed graphs. The model demonstrated very good proficiency in disease prediction, encompassing even rare diseases, for previously unseen patients. In another study, GNNs were used to create robust patient representations from EHRs for chronic disease prediction³¹. The challenges remain, including the trade-off between flexibility and performance in current GNN-based algorithms, as well as the reduction in sample size when introducing multimodal features, which limits the performance of machine learning models²⁹.

Related work

Although the use of EHR data in GNNs for disease classification is increasing, most recent studies on AD classification have focused on imaging modalities or a combination of imaging and other modalities. The use of EHR-based comorbidity data in AD classification, however, has not yet been fully explored. Li et al.³² propose a novel Regional Brain Fusion-Graph Convolutional Network (RBF-GCN) framework for AD prediction and analysis. The RBF-GCN incorporates three neuroimaging modalities: structural MRI, diffusion-weighted imaging (DWI), and amyloid positron emission tomography (PET). The framework is structured into three key modules: the hemispheric network generation module, the multichannel GCN module, and the feature fusion module. The input to the GCN module includes structural brain networks and the output is the representation information which is used to classify the subjects into different stages of AD. This approach captures comprehensive AD-related features through multimodal data integration and considers hemispheric asymmetries, enhancing diagnostic accuracy, while the adaptive node attributes dynamically refine network representation. However, the model's added complexity increases computational demands, and its performance requires validation on larger, more diverse datasets to ensure generalizability beyond the ADNI cohort. In another study, Lin et al.³³ proposed a novel framework combining Convolutional Neural Networks (CNN) and GCN for AD classification, leveraging data from the Alzheimer's Disease Neuroimaging Initiative (ADNI). The CNN, specifically a 3D-DenseNet, was used to extract high-level anatomical features from MRI scans, while the GCN processed population graphs, where nodes represented individual subjects and edges encoded pairwise similarities using both imaging and non-imaging data. The study conducted three experiments to assess the impact of incorporating demographic and neuropsychological data into the edge-assigning function of the GCN on classification performance across different stages of AD. The framework demonstrated strong performance in both binary and multi-class classifications, with accuracies of 91.6% for AD vs. CN, 91.2% for AD vs. MCI, 96.8% for MCI vs. CN, and 89.4% for AD/MCI/CN. In a similar study, a graph reasoning module (GRM) was introduced to enhance CNN-based AD detection models by incorporating GCNs for relation-aware feature extraction³⁴. This module includes an adaptive graph Transformer (AGT) block, a GCN block, and a feature map reconstruction (FMR) block, which work together to transform input feature maps into graph representations, extract features through GCNs, and reconstruct updated feature maps. These studies highlight the effectiveness of integrating CNNs and GNNs for improved AD classification by utilizing both imaging and non-imaging data. However, the method's reliance on complex graph structures and the challenge of over-smoothing in GNNs are noted as potential limitations.

Motivation and research contributions

To address this question and the limitations of current approaches, we introduce a comorbidity-based classification framework utilizing GNNs, designed to accurately identify individuals across multiple classes (CN, MCI, and AD) as well as in binary classification. The motivation for this work stems from the urgent need to explore more accessible and cost-effective alternatives to imaging-based methods, which are often constrained by high costs and limited accessibility. The novelty of this work lies in its innovative use of EHR-derived comorbidity data for GNN-based AD classification, setting it apart from other studies that predominantly rely on imaging data. By incorporating comorbidities into the GNN models, the study broadens the range of input data and enhances the model's predictive capability using more accessible and cost-effective sources. Additionally, the study stands out for its comprehensive evaluation through ablation studies, which rigorously assess the impact of different feature sets, such as demographics, cognitive scores, and comorbidities, on the model's performance. Each patient is represented as a node in the graph, and these features are incorporated as node or edge features, enabling the model to capture complex relationships and improve classification accuracy. This in-depth analysis provides valuable insights into the most critical features for accurate AD classification. The outcomes of our

investigation demonstrate the capability of our proposed model to achieve accurate classification results. Our main contributions can be summarized as follows:

- Incorporating comorbidity data from EHRs into GNN models for AD classification. This method leverages readily available and cost-effective data, eliminating the need for additional data collection time or resources.
- Proposing an AD classification framework using GNNs, which outperforms traditional ML models in handling complex relationships inherent in EHRs.

Organization of the paper

The remainder of this paper is structured as follows: “Results” presents the results, “Discussion” discusses the findings and their implications, “Materials and methods” outlines the proposed methodology, and “Conclusion” concludes the study while suggesting directions for future research.

Results

Study cohort characteristics

The ADNI has three cohorts: CN, MCI, and AD participants. Subject demographics of the three baseline diagnosis states are provided in Table 1. In ADNI, the CN group includes 423 individuals, the MCI group has 667 individuals, and the AD group consists of 241 individuals. Demographically, the percentage of males is highest in the MCI group (57.4%), followed by the AD group (56.4%) and the CN group (49.2%) with a significant p-value of 0.024. The mean age is highest in the AD group (74.5 years) compared to the CN (73.9 years) and MCI (72.9 years) groups, with a p-value of 0.004 indicating statistical significance. Cognitive scores reveal substantial differences among the cohorts. The Mini-Mental State Examination (MMSE) scores decrease significantly from CN (29.0) to MCI (27.6) to AD (23.3), with a p-value of less than 0.001. Similarly, the Clinical Dementia Rating Sum of Boxes (CDRSB) scores escalate from 0.04 in CN to 1.48 in MCI and 4.30 in AD (p-value < 0.001). The Rey Auditory Verbal Learning Test (RAVLT) immediate recall scores and the Logical Memory Delay Total (LDELTOTAL) scores also show significant declines from CN to AD, both with p-values less than 0.001. In terms of comorbidities, cardiovascular conditions are fairly consistent across the cohorts, with 17.8% in CN, 18% in MCI, and 18.9% in AD (p-value = 0.792). However, gastrointestinal issues are more prevalent in the CN group (13.2%) compared to MCI (12.7%) and AD (9.7%), with a significant p-value of 0.028. The prevalence of

	ADNI				AIBL			
	CN	MCI	AD	p-value*	CN	MCI	AD	p-value*
N	423	667	241		609	144	105	
Demographics								
Male (%)	49.2	57.4	56.4	0.024	56	45.1	59	0.039
Age (mean (SD))	73.9 (5.91)	72.9 (7.54)	74.5 (7.73)	0.004	69.5 (8.16)	75.2 (6.88)	83.8 (9.77)	0.188
Cognitive scores								
MMSE (mean (SD))	29.0 (1.18)	27.6 (1.85)	23.3 (2.02)	<0.001	28.7 (1.27)	26.9 (2.18)	20.3 (5.29)	<0.001
CDRSB (mean (SD))	0.04 (0.15)	1.48 (0.87)	4.30 (1.57)	<0.001	0.03 (0.12)	0.47 (0.14)	0.95 (0.51)	<0.001
RAVLT.immediate.bl (mean (SD))	44.9 (9.81)	34.88 (10.95)	23.2 (7.30)	<0.001	12.6 (3.83)	7.35 (3.55)	3.36 (2.73)	<0.001
LDELTOTAL (mean (SD))	11.2 (4.10)	4.65 (3.70)	0.99 (1.79)	<0.001	13.0 (3.30)	6.10 (3.50)	1.40 (1.85)	<0.001
Comorbidities								
Cardiovascular (count (%))	294 (17.8)	476 (18)	169 (18.9)	0.792	260 (42.7)	68 (47.2)	37 (35.2)	0.166
Hepatic (count (%))	15 (0.9)	26 (1.0)	8 (0.9)	0.956	24 (3.9)	7 (4.9)	3 (2.9)	0.725
Musculoskeletal (count (%))	315 (19.1)	468 (17.7)	147 (16.4)	0.222	311 (51.1)	68 (47.2)	37 (35.2)	0.096
Gastrointestinal (count (%))	218 (13.2)	336 (12.7)	87 (9.7)	0.028	165 (27.1)	45 (31.2)	20 (19.0)	0.096
Endocrine-metabolic (count (%))	209 (12.7)	320 (12.1)	119 (13.3)	0.626	114 (18.7)	23 (16.0)	16 (15.2)	0.563
Renal-genitourinary (count (%))	188 (11.4)	308 (11.7)	104 (11.6)	0.967	35 (5.7)	8 (5.6)	0 (0.0)	0.042
Smoking (count (%))	111 (6.7)	191 (7.2)	77 (8.6)	0.217	158 (25.9)	61 (42.4)	36 (34.3)	<0.001
Malignancy (count (%))	120 (7.3)	160 (6.1)	50 (5.6)	0.162	101 (16.6)	24 (16.7)	14 (13.3)	0.696
Neurologic (count (%))	122 (7.4)	237 (9.0)	66 (7.4)	0.117	60 (9.9)	18 (12.5)	37 (35.2)	<0.001
Depressed mood (count (%))	11 (0.7)	28 (1.1)	29 (3.2)	<0.001	567 (94.6)	127 (88.2)	80 (76.2)	<0.001
Nausea (count (%))	2 (0.1)	2 (0.1)	3 (0.3)	0.185	517 (84.9)	116 (80.6)	83 (79)	0.195
Diarrhea (count (%))	8 (0.5)	21 (0.8)	11 (1.2)	0.120	517 (84.9)	116 (80.6)	83 (79)	0.195
Constipation (count (%))	17 (1.0)	34 (1.3)	10 (1.1)	0.740	517 (84.9)	116 (80.6)	83 (79)	0.195
Abdominal discomfort (count (%))	8 (0.5)	20 (0.8)	4 (0.4)	0.421	517 (84.9)	116 (80.6)	83 (79)	0.195
Blurred vision (count (%))	9 (0.5)	13 (0.5)	7 (0.8)	0.600	370 (60.8)	73 (50.7)	46 (43.8)	0.001
Fall (count (%))	2 (0.1)	3 (0.1)	2 (0.2)	0.729	585 (96.1)	135 (93.8)	95 (90.5)	0.04

Table 1. Subject demographics and clinical features at baseline for ADNI and AIBL. *p≤0.05, ANOVA, Kruskal–Wallis test, or x² test, as appropriate.

depressed mood is significantly higher in the AD group (3.2%) compared to CN (0.7%) and MCI (1.1%) with a p-value of less than 0.001. This analysis highlights the significant differences in demographics, cognitive scores, and some comorbidities among the CN, MCI, and AD cohorts within the ADNI dataset, highlighting the progressive nature of Alzheimer's disease and its impact on cognitive and overall health.

Performance evaluation of GNN models

In this work, we present a pipeline for binary and multi-class AD classification using GNNs with EHR data, demographic, and cognitive scores. We performed extensive benchmarking work on how GNNs work with respect to three ML models: random forest (RF), support vector machine (SVM), and extreme gradient boosting (XGBoost). We first discuss binary classification results on the original feature set, followed by multi-class classification, and conclude each part with comparisons to existing literature. The workflow of our GNN-based model for AD classification is presented in Fig. 1. The AD classification task, modeled as a supervised node classification problem, aims to categorize individuals into two (CN/AD, CN/MCI, AD/MCI) or three (CN, MCI, AD) categories. The proposed GNN models were trained and tested as detailed in the methodology section, with performance results summarized in Table 5.

Binary classification

In the first binary classification task (CN vs. AD), the graph convolution network (GCN) and ChebConv models performed exceptionally well. Both models achieved an accuracy (ACC) of 0.99, an AUC of 0.99, and an F1-score of 0.99. The graph attention network (GAT) model, on the other hand, showed significantly lower performance with metrics of (ACC = 0.56, AUC = 0.49, F1-score = 0.41). The graph sample and aggregate (SAGE) model also had lower performance, with metrics of (ACC = 0.45, AUC = 0.51, F1-score = 0.43). The traditional ML

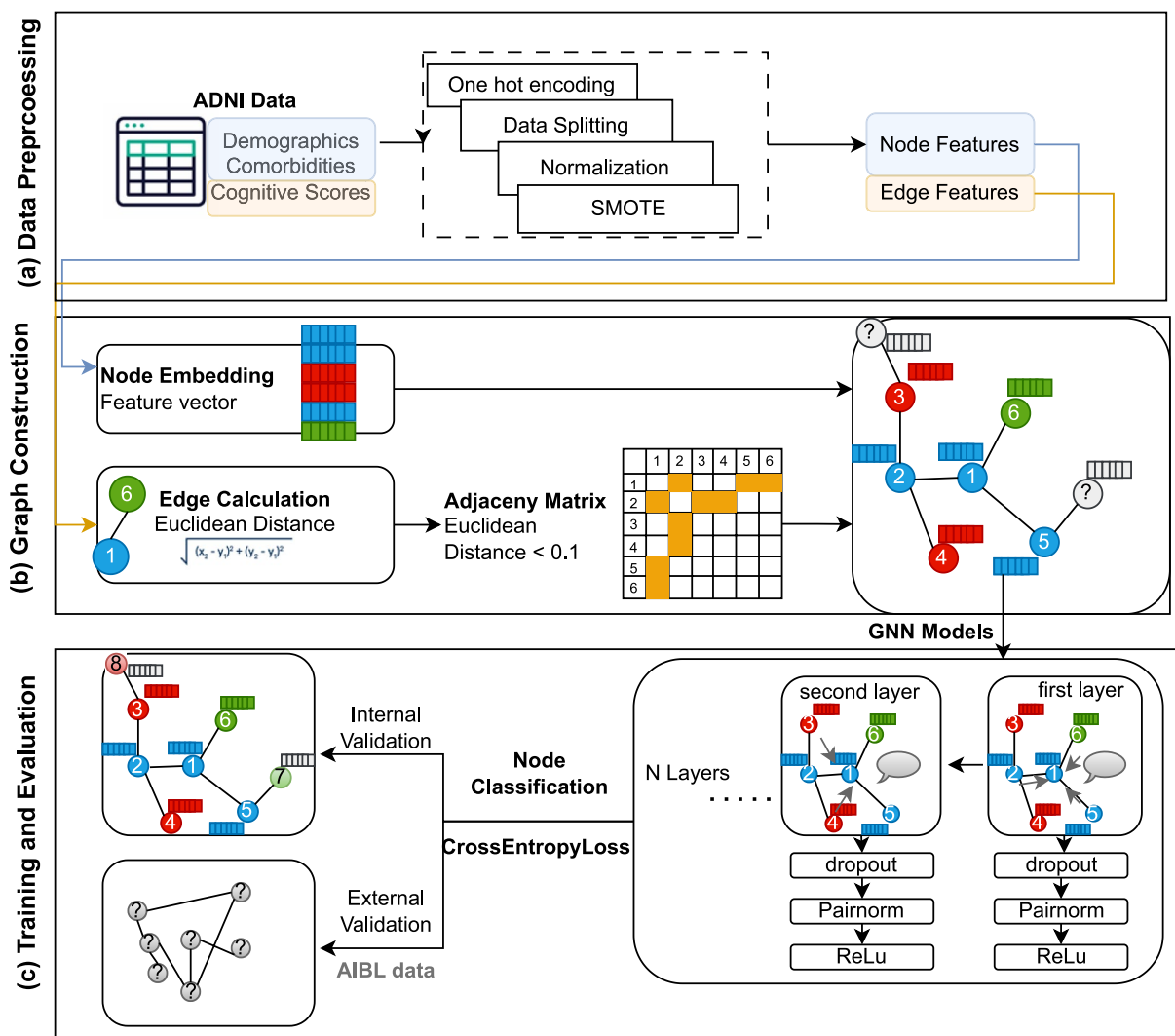


Figure 1. Overview of the system model for Alzheimer's disease classification using graph neural networks. The model consists of three key stages: (a) data preprocessing, (b) graph construction, and (c) GNN model training and evaluation. In (b), patients are represented as nodes with edges connecting them based on cognitive score similarities. In (c), the trained models are validated on test set and also externally using the AIBL dataset.

models, including SVM, RF, and XGBoost, showed lower performance compared to the GNN models. SVM achieved (ACC = 0.59, AUC = 0.57, F1-score = 0.55), while RF and XGBoost recorded similar metrics with ACC around 0.57–0.55, AUC around 0.56–0.52, and F1-scores around 0.57–0.55. The GNN models were superior, particularly GCN and ChebConv, to effectively distinguish between CN and AD individuals, as evidenced by their significantly higher performance metrics.

The t-SNE visualization provides valuable insights into the distribution of nodes for the binary classification task using original features before and after the training process, Fig. 3a, b respectively. Before training, nodes are scattered in a high-dimensional space based on their initial features. As the ChebConv undergoes training, the t-SNE visualization shows the model's ability to cluster nodes in a lower-dimensional space, highlighting the network's capacity to capture meaningful relationships between nodes. The visualization post-training demonstrates a more structured arrangement of nodes, reflecting the enhanced representation learning achieved by the ChebConv. Figure 3c presents the confusion matrix for the ChebConv, while Fig. 3d illustrates the accuracy plots of the four models. In the second binary classification task (MCI vs. AD) the ChebConv model achieves highest performance (ACC = 0.93, AUC = 0.96, F1-score = 0.93), indicating continued robustness. The t-SNE visualization shows the distribution of nodes for the MCI vs. AD binary classification task using original features before and after the training process, Fig. 3e, f respectively. The GCN model also performed well with metrics of (ACC = 0.84, AUC = 0.95, F1-score = 0.84). The GAT and SAGE models again showed lower performance, with the GAT model achieving (ACC = 0.50, AUC = 0.52, F1-score = 0.37), while the SAGE model had metrics of (ACC = 0.51, AUC = 0.57, F1-score = 0.50). Interestingly, the traditional ML models, such as SVM, RF, and XGBoost, performed better than GAT and SAGE but still fell short of the ChebConv model, highlighting the latter's superior effectiveness. This further indicates that the GAT and SAGE models are less effective in capturing the distinctions between AD and MCI compared to the GCN and ChebConv models (Fig. 3g presents the confusion matrix for the ChebConv in the MCI vs. AD task). Also seen in the accuracy plots in Fig. 3h. In the third binary classification task (CN vs. MCI), the ChebConv model again led the performance with metrics of (ACC = 0.78, AUC = 0.83, F1-score = 0.76). The GCN model followed with metrics of (ACC = 0.65, AUC = 0.75, F1-score = 0.64). These results show that while the ChebConv model maintains high effectiveness, the task of distinguishing between CN and MCI, which is crucial for early detection of cognitive decline, is generally more challenging, as reflected in the lower performance metrics compared to the CN vs. AD classification. The GAT and SAGE models had the lowest performance in this task as well. The GAT model achieved (ACC = 0.54, AUC = 0.47, F1-score = 0.39). The SAGE model recorded metrics of (ACC = 0.52, AUC = 0.50, F1-score = 0.47).

Multi-class classification

In the multi-class classification task, the ChebConv model achieves the highest accuracy (ACC = 0.98, AUC = 1.0, F1-score = 0.98) when all features, as described in the original feature set, are used. The performance can be attributed to the Chebyshev polynomials used in the convolution operation, which likely enhance the model's ability to capture relevant features and complex relationships within the data, leading to superior classification accuracy and robustness. The GAT model achieved an (ACC = 0.93, AUC = 0.99, F1-score = 0.93). While its performance is slightly lower than that of the ChebConv model, it still demonstrates strong classification capabilities. The SAGE model attained an (ACC = 0.83, AUC = 0.94, F1-score = 0.83). The SAGE model's approach to aggregating information from neighboring nodes is beneficial but might not be as effective as the specialized convolution and attention mechanisms used in the other models. Lastly, the GCN model which exhibited the lowest performance among the evaluated models, with (ACC = 0.81, AUC = 0.92, F1-score = 0.81). In comparison, traditional ML models (SVM, RF, and XGBoost) performed lower in the multi-class classification task, with ACC ranging from 0.37 to 0.44, AUC from 0.57 to 0.62, and F1-scores from 0.38 to 0.43, further highlighting the superior performance of the GNN models, particularly ChebConv. In conclusion, the ChebConv model outperforms other models in the multi-class classification task, achieving the highest accuracy, AUC, and F1-score. The high performance of the ChebConv in the multi-class classification task is reflected in the confusion matrix as seen in Fig. 4c. The diagonal elements indicate high percentages for each class with minimal off-diagonal values suggesting effective classification across all classes. The t-SNE visualization provides valuable insights into the distribution of nodes before and after the training process, Fig. 4a, b respectively. Before training, nodes are scattered in a high-dimensional space based on their initial features. As the ChebConv undergoes training, the t-SNE visualization shows the model's ability to cluster nodes in a lower-dimensional space, highlighting the network's capacity to capture meaningful relationships between nodes. The visualization post-training demonstrates a more structured arrangement of nodes, reflecting the enhanced representation learning achieved by the ChebConv. In Fig. 4d, the AUC-ROC evaluation metrics are displayed with the 95% confidence intervals shown in grey. Figure 4e illustrates the accuracy intervals for all the GNN models, clearly demonstrating that the ChebConv model is performing the best. Finally, Fig. 4f presents the F1-scores for all the GNN models.

While no study has employed the same data modality for identical classification purposes, our work, which encompasses non-imaging data (demographic information, cognitive scores, and comorbidities), can be compared to the top studies which performed 3-way classification on ADNI data as seen in Table 2. Our work achieved a remarkable multi-class classification accuracy of 98%. This surpasses the results reported in the literature for multi-class classification using GNNs³², which relied on imaging data such as sMRI, PET, and DWI information and achieved an accuracy of 90%. The strength of our original feature set lies in its ability to deliver superior performance without the need for imaging data, underscoring its effectiveness and versatility in clinical applications. In the available literature, varying data quantities and preprocessing methods were used, making direct comparisons potentially unjustified or unreasonable. Nonetheless, such a comparison offers valuable references for the academic community.

Ablation study results

Binary classification

In Ablation Study 1, where comorbidities were removed, the models showed varying degrees of performance changes. The GCN and ChebConv models maintained high performance in the AD vs. CN task with metrics of (ACC = 0.99, AUC = 0.99, F1-score = 0.99). However, the GAT and SAGE models showed slight improvements, achieving (ACC = 0.58, AUC = 0.53, F1-score = 0.44) and (ACC = 0.50, AUC = 0.53, F1-score = 0.40) respectively. For the AD vs. MCI task, the ChebConv model's performance slightly decreased to (ACC = 0.92, AUC = 0.96, F1-score = 0.92), while GCN improved to (ACC = 0.87, AUC = 0.95, F1-score = 0.87). The GAT model showed a minor improvement with metrics of (ACC = 0.50, AUC = 0.49, F1-score = 0.37), and SAGE also improved to (ACC = 0.53, AUC = 0.53, F1-score = 0.45). In the CN vs. MCI classification task, ChebConv's performance increased to (ACC = 0.85, AUC = 0.90, F1-score = 0.85), while GCN also improved to (ACC = 0.68, AUC = 0.75, F1-score = 0.67). The GAT model showed a slight improvement (ACC = 0.48, AUC = 0.52, F1-score = 0.32), and SAGE improved to (ACC = 0.51, AUC = 0.55, F1-score = 0.39). Throughout Ablation 1, the ML models performed similarly to GAT and SAGE, but significantly lower than ChebConv. Overall, the removal of comorbidities in Ablation Study 1 showed the importance of comorbidities in enhancing classification accuracy.

In Ablation Study 2, where comorbidities and cognitive scores were switched, the GCN model's performance significantly dropped in the AD vs. CN task to (ACC = 0.46, AUC = 0.47, F1-score = 0.45), while the ChebConv model showed a slight decrease to (ACC = 0.99, AUC = 0.99, F1-score = 0.99). The GAT model improved to (ACC = 0.59, AUC = 0.64, F1-score = 0.51), and SAGE showed significant improvement to (ACC = 0.98, AUC = 0.99, F1-score = 0.98). For the AD vs. MCI task, the GCN model's performance dropped significantly to (ACC = 0.31, AUC = 0.54, F1-score = 0.21). The ChebConv model showed a slight decrease to (ACC = 0.93, AUC = 0.98, F1-score = 0.93). The GAT and SAGE models showed improved performance with (ACC = 0.60, AUC = 0.53, F1-score = 0.52) and (ACC = 0.87, AUC = 0.97, F1-score = 0.87) respectively. In the CN vs. MCI classification task, ChebConv's performance increased significantly to (ACC = 0.94, AUC = 0.99, F1-score = 0.94) giving the highest performance across all feature sets in this task. The t-SNE visualization provides insight into the distribution of nodes for the CN vs. MCI binary classification task using ablation 2 features before and after the training process, Fig. 3i, j respectively. GCN's performance dropped to (ACC = 0.42, AUC = 0.50, F1-score = 0.34), while GAT showed an improvement to (ACC = 0.53, AUC = 0.51, F1-score = 0.44), and SAGE improved to (ACC = 0.88, AUC = 0.99, F1-score = 0.89). The accuracy plots for the four models is shown in Fig. 3l, while Fig. 3k presents the confusion matrix for the best performing model, ChebConv. Here, the ML models matched the performance of ChebConv, likely because they have more direct access to node information rather than relying on connections between nodes (edges). The restructured features may have amplified the importance of individual node attributes, which traditional ML models are better equipped to exploit. Overall, switching comorbidities and cognitive scores in Ablation Study 2 highlighted the dependency of the GCN model on the original feature structure, resulting in a significant performance drop. However, the ChebConv model remained robust, while the GAT and SAGE models showed notable improvements, indicating their flexibility in handling different feature structures.

In Ablation Study 3, where cognitive scores were removed, all GNN models experienced a significant drop in performance. The GCN model's metrics for the AD vs. CN task were (ACC = 0.46, AUC = 0.47, F1-score = 0.44), and the ChebConv model dropped to (ACC = 0.56, AUC = 0.55, F1-score = 0.56). The GAT model showed a slight improvement with (ACC = 0.52, AUC = 0.49, F1-score = 0.42), and SAGE had (ACC = 0.49, AUC = 0.47, F1-score = 0.40). In this case, the ML models outperformed the GNN models, with the RF model achieving the best results (ACC = 0.58, AUC = 0.61, F1-score = 0.58). For the AD vs. MCI task, the GCN model showed a significant decrease to (ACC = 0.29, AUC = 0.54, F1-score = 0.16). The ChebConv model dropped to (ACC = 0.46, AUC = 0.61, F1-score = 0.46). The GAT model had (ACC = 0.59, AUC = 0.53, F1-score = 0.51), and the SAGE model showed (ACC = 0.44, AUC = 0.53, F1-score = 0.39). In the CN vs. MCI classification task, GCN had (ACC = 0.45, AUC = 0.50, F1-score = 0.35), ChebConv dropped to (ACC = 0.53, AUC = 0.60, F1-score = 0.51), GAT showed (ACC = 0.52, AUC = 0.51, F1-score = 0.39), and SAGE had (ACC = 0.52, AUC = 0.59, F1-score = 0.44). Again, the ML models outperformed the GNN models, where the SVM model has the best results (ACC = 0.56, AUC = 0.62, F1-score = 0.57). The removal of cognitive scores in Ablation Study 3 resulted in the most significant performance decline across all models and tasks. This highlights the crucial role of cognitive scores in classification tasks, indicating that these features are essential for accurate classification in distinguishing between different cognitive states.

Table 2 provides a performance comparison of our best performing models in different tasks. For the CN vs. AD task, this work achieves an impressive accuracy of 99%, significantly outperforming the closest study from 2022³², which reported an accuracy of 96%. In the CN vs. MCI task, while the 2023 study³³ achieved a slightly higher accuracy of 96.8%, this work still achieved a strong accuracy of 94% without utilizing any imaging modalities. For the MCI vs. AD task, the 2022 study³² reported the highest accuracy of 95%, whereas this work achieved a commendable accuracy of 93%. Also, in the study by Grassi et al. (2019)³⁵, a weighted rank average ensemble method, including logistic regression (LR) and other machine learning techniques, was used to classify MCI vs. AD. This study, which utilized demographic and cognitive data, achieved an accuracy of 79%. Despite employing an ensemble of multiple machine learning algorithms, the results were lower. In contrast, this work utilized GNNs (ChebConv) and incorporated demographic, cognitive, and clinical data. The classification tasks performed in this work yielded significantly higher accuracy, 93% for MCI vs. AD. This comparison highlights the importance of incorporating clinical information into the models. Despite using the same dataset, the inclusion of clinical data in this work led to substantially higher performance, even surpassing the results achieved by the ensemble machine learning techniques used in the 2019 study.

Additionally, we conducted a loss comparison specifically for the early detection classification task (CN vs. MCI) across the different ablation studies. This analysis focused on the top-performing model throughout the ablations, which is the ChebConv model. The loss comparison is presented in Fig. 5, where we analyze the impact of systematically varying the feature set—including the removal of comorbidities (Ablation 1, Fig. 5b), switching comorbidities with cognitive scores (Ablation 2, Fig. 5c), and removing cognitive scores (Ablation 3, Fig. 5d)—on the model's ability to minimize error during training and testing. In Fig. 5a, with the original feature set, the training and validation losses closely follow each other, indicating strong generalization and convergence. In Fig. 5b (Ablation 1), there's a mid-training spike in validation loss, suggesting instability due to the removal of comorbidities. Figure 5c (Ablation 2) shows a more stable loss curve, but the training and validation losses are less aligned compared to the original, indicating less effective learning. This analysis shows that the graph configuration, particularly feature structure and integration, does impact performance. The varying loss values across ablations highlight the crucial role of certain features, like cognitive scores, in achieving high accuracy. The ChebConv model's robustness is closely linked to its effective use of these features, leading to superior performance and demonstrating the importance of careful feature selection and representation.

Multi-class classification

In the first ablation study, where comorbidities were removed, all the models exhibited a decrease in performance. The ChebConv model's performance decreased to (ACC = 0.93, AUC = 0.99, F1-score = 0.93). The GAT model's performance also dropped to (ACC = 0.91, AUC = 0.98, F1-score = 0.91). The GCN model showed a significant decrease, with (ACC = 0.51, AUC = 0.75, F1-score = 0.48). The SAGE model had a performance of (ACC of 0.64, AUC = 0.86, F1-score = 0.63). In this part, where comorbidities were removed, the ML models (SVM: ACC = 0.36, AUC = 0.54, F1 = 0.38; RF: ACC = 0.35, AUC = 0.53, F1 = 0.36; XGBoost: ACC = 0.32, AUC = 0.50, F1 = 0.33) showed a decrease in performance, and the GNN models outperformed them significantly. Removing comorbidities led to a noticeable decline in the models' performance which suggests that comorbidity information may play a crucial role in accurately classifying different cognitive states.

In the second ablation study, where comorbidities and cognitive scores were switched, the ChebConv model performed well with an (ACC = 0.95, AUC = 0.99, F1-score = 0.95). The GAT model achieved an (ACC = 0.94, AUC = 0.99, F1-score = 0.94). The GCN model had an (ACC = 0.63, AUC = 0.84, F1-score = 0.55). The SAGE model showed an (ACC = 0.65, AUC = 0.83, F1-score = 0.59). Here, the ML models (SVM: ACC = 0.91, AUC = 0.97, F1 = 0.91; RF: ACC = 0.94, AUC = 0.99, F1 = 0.94; XGBoost: ACC = 0.93, AUC = 0.99, F1 = 0.93) experienced a sharp increase in performance comparing to GNN models. Switching comorbidities and cognitive scores did not significantly affect the ChebConv and GAT models, suggesting their robustness in handling feature modifications. However, the GCN and SAGE models experienced a moderate decline, indicating their reliance on the original feature set's structure. The increase seen in ML methods could be attributed to the cognitive scores being part of the node feature input of the ML.

In the third ablation study, where cognitive scores were removed, the ChebConv model's performance dropped significantly to an (AUC = 0.60, ACC = 0.42, F1-score = 0.42). The GAT model also showed a notable decline, with an (AUC = 0.59, ACC = 0.40, F1-score = 0.40). The GCN model had an (AUC = 0.57, ACC = 0.45, F1-score = 0.38). The results where cognitive scores were removed ML (SVM: ACC = 0.36, AUC = 0.54, F1 = 0.38; RF: ACC = 0.35, AUC = 0.53, F1 = 0.36; XGBoost: ACC = 0.32, AUC = 0.50, F1 = 0.33), experienced a significant performance drop just like all models. The SAGE model showed the highest score of an (AUC = 0.65, ACC = 0.47, F1-score = 0.34). The removal of cognitive scores led to the most significant performance decline across all models which is unsurprising, as cognitive scores are recognized as crucial determinants, thus reaffirming the effectiveness of the models.

Model interpretation and feature analysis

Model interpretation of GNNs can be achieved through node feature importance analysis. By assessing the contribution of each node feature to the model's predictions, we can understand which features are most influential. Such an analysis helps in uncovering the decision-making process of GNNs, providing insights into the model's behavior. We examine the top 5 features selected by the top-performing model in multi-class classification (ChebConv), which assigns different features for each of the three classes: CN, MCI, and AD. In our analysis, we examine the features selected by the model for the three classes using the original feature set, as this provided the highest performance. Additionally, since the ChebConv model using the original feature set mostly achieved the highest performance in binary classification, we will also examine the features selected by this model across the three binary classification tasks. This comprehensive approach allows us to understand the model's behavior and feature importance in various contexts.

When examining the features identified by the top-performing ChebConv model in the multi-class classification, certain node features—(1) Age, (2) Diarrhea, (3) Hepatic, (4) Nausea, and (5) Abdominal discomfort—demonstrate varying levels of importance across the three classes CN, MCI, and AD, as seen in Fig. 7a–c respectively. “AGE” has a mean importance of 4.15 in CN, 3.79 in MCI, and 2.97 in AD, indicating that while age is an important factor across all groups, its impact diminishes slightly in those who progress to AD. “Diarrhea” shows consistent importance with means of 3.42 in CN, 3.44 in MCI, and 2.51 in AD, suggesting it may be a more significant issue in the earlier stages of cognitive decline. “Hepatic” issues are fairly stable, with means of 3.59 in CN, 3.40 in MCI, and 2.54 in AD, emphasizing the importance of liver health in maintaining metabolic balance and reducing neuroinflammation, which is implicated in AD pathogenesis³⁶. “Nausea” remains relevant across all classes, with mean importances of 3.36 in CN, 3.14 in MCI, and 2.23 in AD, likely linked to medication use or underlying health conditions common in the elderly³⁷. “Abdominal discomfort” also features prominently, with mean importances of 3.36 in CN, 3.08 in MCI, and 2.69 in AD, pointing to gastrointestinal issues as a consistent

concern that could influence or be influenced by cognitive health. These comparisons highlight how certain features vary in their significance across the stages of cognitive decline, providing new insights that warrant a closer examination of factors like abdominal discomfort and diarrhea as potential markers for distinguishing between CN, MCI, and AD. This emphasizes the interconnectedness of physical and cognitive health in the progression of Alzheimer's Disease, suggesting that these physical symptoms could play a more critical role in understanding and potentially intervening in the early stages of cognitive decline.

Next, we analyze the top features selected by the ChebConv model using the original feature set for the three binary classification tasks, as shown in Fig. 6. The mean importance of features across the cross-validation runs provides insights into their significance for each class. In the CN vs. AD classification (Fig. 6a), the feature "Gender Female" stands out with a mean importance of 0.20, showing there is a higher risk of AD in women in this specific cohort, which is often linked to longer life expectancy and hormonal changes. Other key features include "Diarrhea" (mean importance of 0.19) and "Constipation" (mean importance of 0.18), both of which are common gastrointestinal issues that can be exacerbated by cognitive decline and medications. "Renal-Genitourinary" (mean importance of 0.18) and "Abdominal Discomfort" (mean importance of 0.18) further emphasize the significance of gastrointestinal and renal health in distinguishing between CN and AD. For the AD vs. MCI classification (Fig. 6b), the top features reflect the progression of AD and its impact on overall health. "Nausea" (mean importance of 2.43) is the most significant feature, followed by "Diarrhea" (mean importance of 1.63), both indicating the increasing influence of autonomic nervous system dysfunction. "Hepatic" (mean importance of 1.50) and "Constipation" (mean importance of 1.47) continue to be important, pointing to their roles in advanced cognitive impairment and overall health deterioration. "Smoking" (mean importance of 1.40) also appears as an important factor in this classification. In the CN vs. MCI classification (Fig. 6c), the model identifies "Fall" (mean importance of 8.08) and "Age" (mean importance of 7.63) as the most crucial features, highlighting the association between motor function decline, aging, and early cognitive impairment. "Constipation" (mean importance of 5.94) and "Depressed Mood" (mean importance of 4.23) also play significant roles, indicating the importance of gastrointestinal health and psychological factors in the transition from CN to MCI. Finally, "Hepatic" (mean importance of 4.11) remains a key feature, underscoring its relevance in overall metabolic health and cognitive function. Overall, these patterns emphasize similar to the in the multi-class classification that gastrointestinal and hepatic health, neurological symptoms, psychological factors, and aging are critical in the progression from normal cognition to MCI and AD. These insights are valuable for early detection and intervention strategies.

External validation

The Australian Imaging, Biomarker & Lifestyle Flagship Study of Ageing (AIBL)³⁸ dataset serves as an external validation dataset for the models trained on ADNI, providing an independent cohort to assess the robustness and generalizability of the models. Similar to ADNI, AIBL has three cohorts: CN, MCI, and AD participants. Subject demographics of the three baseline diagnosis states are provided in Table 1. The AIBL dataset includes 609 CN, 144 MCI, and 105 AD individuals. Compared to ADNI, AIBL shows similar but more pronounced declines in cognitive scores across cohorts. The MMSE scores decrease from CN (28.7) to MCI (26.9) to AD (20.3), with a p-value of less than 0.001. The CDRSB scores differ noticeably between datasets. In ADNI, scores range widely from 0.04 in CN to 4.30 in AD, while in AIBL scores are lower and more compressed, ranging from 0.03 in CN to 0.95 in AD. This narrower range suggests possible differences in population characteristics or assessment criteria. In terms of comorbidities, cardiovascular conditions are more prevalent in the AIBL CN group (42.7%) compared to ADNI (17.8%). The prevalence of depressed mood is markedly higher in AIBL across all groups, with 94.6% in CN, 88.2% in MCI, and 76.2% in AD, all with a significant p-value of less than 0.001. These differences between AIBL and ADNI could be attributed to variations in data collection methods, population characteristics, or healthcare systems in the respective communities. As a result, these discrepancies may affect the generalizability of GNN models trained on one dataset when applied to another, potentially leading to reduced model performance when transferred between datasets.

The top-performing model in each classification task, trained on ADNI data, was used for external validation on the AIBL dataset. The results, shown in Table 3, indicate that the GNN models performed well across different classification tasks. The ChebConv model achieved a solid accuracy of 0.89 for CN vs. AD classification using the original feature set. However, the performance was more modest for MCI vs. AD classification, with an accuracy of 0.64, and for the multi-class classification task (CN vs. MCI vs. AD), with an accuracy of 0.74. The model did perform relatively well in the CN vs. MCI task, achieving an accuracy of 0.84 using the ablation 2 feature set. These results suggest that while the models delivered good outcomes, they did not achieve top-tier performance, which could be due to the differences between the ADNI and AIBL datasets. Nevertheless, the GNN models still demonstrated the ability to produce reasonably accurate predictions when relying solely on comorbidity, cognitive scores, and demographic data. This shows the potential of these models for AD classification using more accessible data types, while also highlighting the challenges posed by dataset variability.

Discussion

The advancements in GNNs and the increasing availability of high-dimensional EHRs have opened new directions in research, particularly in the early prediction of complex diseases like Alzheimer's. Therefore, in this work, we explore how GNNs can effectively utilize EHR-derived data for robust and accurate classification of AD across various stages. We conducted comprehensive benchmarking comparing the performance GNN models to traditional ML models, and utilizing four different input features for binary and multi-class classification tasks. Binary classification is important for prodromal detection, the early identification of individuals at high risk of developing Alzheimer's disease. Detecting MCI as an intermediate stage between normal cognition and AD allows for timely intervention and therapeutic strategies to potentially slow down or mitigate further cognitive

decline. Performing multi-class classification is important in understanding the progression and stages of cognitive decline. These categories represent a spectrum of cognitive states, ranging from CN to MCI and finally to dementia by AD. By accurately classifying individuals into these categories, it becomes possible to identify and track the progression of cognitive impairment and disease.

Across all classification tasks, several key parameters and mechanisms were identified as significantly impacting performance: (1) use of EHR-derived comorbidity data, (2) graph representation of patients: by representing each patient as a node within a graph, and utilizing comorbidities, demographics, and cognitive scores as node and edge features, our model effectively captures intricate relationships between patients. (3) ChebConv: The choice of this model is particularly impactful as it utilizes Chebyshev polynomials in the convolution process, which enhances its ability to capture relevant features and complex relationships within the data. This is a key factor in achieving the high accuracy of 0.99% in our classification tasks with the original feature set. (4) Comprehensive ablation studies: our comprehensive ablation studies confirm that the proposed feature set and graph construction are optimal for maximizing classification accuracy.

In general, the performance across different tasks and models was best when the original feature set, which includes comorbidity information, was used. Utilizing comorbidities holds the potential to identify risk factors for future development of Alzheimer's disease in individuals currently classified as normal. Additionally, although the treatment of these comorbidities will not undo the cognitive dysfunction, they should be recognized as co-factors and treated immediately. This information could prove valuable in clinical settings, allowing for a focus on preventive care based on patients' medical history before resorting to costly modalities like MRI and PET data.

Across all three binary classification tasks (AD vs. CN, AD vs. MCI, CN vs. MCI), the ChebConv model consistently outperforms other GNN models, with AUC values ranging from 0.99 to 0.98, accuracies ranging from 0.99 to 0.93, and F1 scores ranging from 0.99 to 0.93, demonstrating its robust ability to handle complex graph-structured data. The GCN model also showed strong performance, particularly in the AD vs. CN task, but exhibited a significant dependency on the original feature structure, as evidenced by performance drops in ablation studies. The GAT and SAGE models, along with the ML models, on the other hand, consistently underperformed, which could be attributed to their limitations in handling this specific intricate graph-structured data. Although using the original feature set consistently provided the best overall performance, the ablation studies provided valuable insights into the importance of different features. The removal of comorbidities generally led to a slight decline in the performance of the ChebConv and GCN models, emphasizing the importance of these features. Interestingly, the GAT and SAGE models showed slight improvements, suggesting these models might benefit from focusing on other features. The most significant performance decline across all models occurred when cognitive scores were removed, highlighting their critical role in accurate classification. Notably, ML models performed better or close to GNNs in Ablation 3, with identical results for Ablation 1 and Ablation 3 due to the same input (node features) being used in both scenarios. Comparatively, our GNN models, particularly ChebConv, demonstrated higher accuracy than recent studies, emphasizing the effectiveness of incorporating a comprehensive feature set, with demographic, cognitive, and clinical data. Overall, the ChebConv model's robustness and effectiveness, especially when using the full feature set, highlight its potential for early and accurate prediction of Alzheimer's disease progression, with comorbidities playing a significant role in enhancing classification accuracy.

A similar pattern was seen in the multi-class classification tasks, where the ChebConv model consistently outperformed other GNN models when using original feature set. This could be attributed to the Chebyshev polynomials used in its convolution operation, enhancing its ability to capture complex relationships within the data. The GAT model, while performing slightly lower than ChebConv, still showed strong classification capabilities, indicating its effectiveness in handling diverse data structures. The SAGE model, although performing well, lagged behind ChebConv and GAT, suggesting that its aggregation approach might not be as effective as the specialized convolution and attention mechanisms used in the other models. The GCN model exhibited the lowest performance, reflecting its dependency on the original feature structure and its limitations in handling high-dimensional EHR data. Even in multi-class classification, when comorbidities were removed, all models exhibited a decrease in performance, particularly the GCN and SAGE models. This suggests that comorbidity information plays a crucial role in accurately classifying different cognitive states and identifying potential risk factors for AD progression. The second ablation study, where comorbidities and cognitive scores were switched, showed that ChebConv and GAT models remained robust, whereas GCN and SAGE models experienced moderate declines, indicating their reliance on the original feature set's structure. The third ablation study, where cognitive scores were removed, resulted in the most significant performance decline across all models, highlighting the critical role of cognitive scores in accurate classification.

Comparing multi-class and binary classification tasks, a general pattern emerged: models performed better when utilizing the original feature set, emphasizing the importance of including comprehensive feature sets for accurate classification. In both classification tasks, the ChebConv model consistently demonstrated robustness and effectiveness, particularly when using the full feature set, underscoring its potential for early and accurate prediction of Alzheimer's disease progression. The clinical significance of these findings lies in the potential to identify risk factors and provide preventive care based on patients' medical history before resorting to costly modalities like MRI and PET data. The technical significance highlights the effectiveness of ChebConv in capturing complex relationships within high-dimensional EHR data, setting a new standard for using GNNs in medical research.

Our models have demonstrated superior performance compared to previously published studies. For the CN vs. AD task, our approach achieved an impressive accuracy, significantly outperforming the closest study from 2022³². Furthermore, in multi-class classification, our work, with an accuracy of 98%, surpassed the results reported in the literature for multi-class classification using GNNs that relied on imaging data³². This highlights the effectiveness of incorporating a comprehensive feature set, including demographic, cognitive, and clinical

data, into our models, emphasizing their potential for early and accurate predictions of Alzheimer's disease progression.

By examining node feature importance, we identified key factors such as gastrointestinal health, hepatic function, neurological symptoms, depression, and cardiovascular health as pivotal in the progression from CN to MCI and AD. The top-performing ChebConv model revealed that features like "AGE," "Diarrhea," "Hepatic," "Nausea," and "Abdominal Discomfort" showed varying levels of importance across the three classes, with "AGE" having a mean importance of 4.15 in CN, 3.79 in MCI, and 2.97 in AD. This analysis underscores the interconnectedness of physical and cognitive health in the progression of Alzheimer's Disease and highlights the need for early detection and intervention strategies based on these insights. Future work will involve refining and testing the model on other data types, such as genetic information and longitudinal health records, to enhance the model's comprehensiveness and predictive power. Moreover, while the identified biomarkers are promising, they require experimental validation to confirm their clinical relevance.

Lastly, the AIBL dataset was used for external validation to assess the robustness and generalizability of the GNN models trained on ADNI data. Although AIBL shares similarities with ADNI, it exhibits more pronounced declines in cognitive scores and notable differences in comorbidities, such as higher prevalence of cardiovascular conditions and depressed mood. These discrepancies could affect the generalizability of models between datasets. When tested on AIBL, the top-performing ChebConv model achieved good accuracy in the CN vs. AD classification (0.89) but showed more modest performance in MCI vs. AD (0.64) and multi-class classification (0.74). The model performed relatively well in the CN vs. MCI task (0.84), indicating that while the GNN models demonstrated reasonable accuracy using comorbidity, cognitive scores, and demographic data, their performance was impacted by differences between the datasets. These findings underscore the potential of using more accessible data types for AD classification, but also highlight the challenges of dataset variability.

Material and methods

In this section, we outline the approach to enhancing AD classification using GNNs by utilizing readily available data sources. We present the datasets used and describe the data preparation process, followed by the architecture of the proposed Graph Neural Network models designed for AD classification. The overall system model, as depicted in Fig. 1, is organized into three primary stages: Data Preprocessing, Graph Construction, and GNN Model Training and Evaluation. While this overview highlights the key steps, it is intended to give a broad understanding rather than a detailed, step-by-step guide.

Dataset

The data used for this study was derived from the publicly available Alzheimer's Disease Neuroimaging Initiative³⁹ and the Australian Imaging, Biomarker & Lifestyle Flagship Study of Ageing³⁸. Participants were filtered to include all 1331 ADNI participants and 858 AIBL participants with existing EHR data. In ADNI, each participant had four diagnosis states (CN, Early MCI, Late MCI, and AD). To standardize, Early MCI and Late MCI were categorized into MCI. AIBL, which adopted key elements from ADNI, provided three diagnosis states (CN, MCI, AD). Subject demographics for the three baseline diagnosis states are provided in Table 1. For this study, we utilized the "ADNIMERGE"⁴⁰ dataset, which contain demographics and cognitive scores, along with the EHR-driven datasets: "RECMHIST," containing medical history information, and "RECBLLLOG," capturing symptoms data, as depicted in Fig. 2. Similarly, AIBL provides comparable data modalities. Overall, the features extracted from these datasets fall into three main categories: demographic (2 features), cognitive scores (4 features), and medical history (comorbidities—10 features), along with symptoms (6 features). Symptoms persisting over multiple visits were selectively chosen and merged with comorbidities for subsequent analysis. Henceforth in this paper, both symptoms and comorbidities are collectively referred to as "comorbidities".

Data preprocessing

The following section outlines the steps for data preparation, as illustrated in Fig. 2. Following the integration and feature selection described earlier, the final dataset comprise both numeric and categorical data types. One-hot encoding is performed on all categorical features. The dataset is divided into training and testing sets in an 80–20 stratified split. This division ensures that a significant portion of the data was allocated for model training, allowing for robust learning, while the remaining subset served as an independent evaluation set for assessing the model's performance. The AIBL dataset, when used for external validation, was treated similarly to the test set, ensuring that the model's generalizability was properly assessed. The dataset with selected features is free of any missing values, there no imputation needed. Following that, Min-Max scaling is applied to normalize the data. Normalization is executed on the training set and subsequently applied on the test set using the prediction matrix derived from the training set. Finally, to address imbalances in class distribution, Synthetic Minority Over-sampling Technique (SMOTE) is applied to the training data, wherein existing minority class samples are interpolated to generate synthetic samples⁴¹.

At this stage, the training and testing sets are partitioned into edge and node features to compute the adjacency matrix from edge features. An adjacency matrix is a square matrix indicating the presence of edges between nodes in a graph⁴². Entries are set to 1 if there is a direct edge between nodes and 0 otherwise. It encodes graph structure and connectivity for graph-based algorithms. In our graph, we utilize two sets of edge features: cognitive score-based and comorbidity-based. The adjacency matrix is computed from the Euclidean distance from the difference between their cognitive score vectors or the common count of comorbidities, serving as a proxy for edge connections. The calculation of the adjacency matrix is done only with the training set, and the test set is masked to prevent data leakage. The threshold which determines whether an edge is present (1) or not (0) in these distance matrices is considered a hyper parameter. This selected cut-off, a hyperparameter, was determined

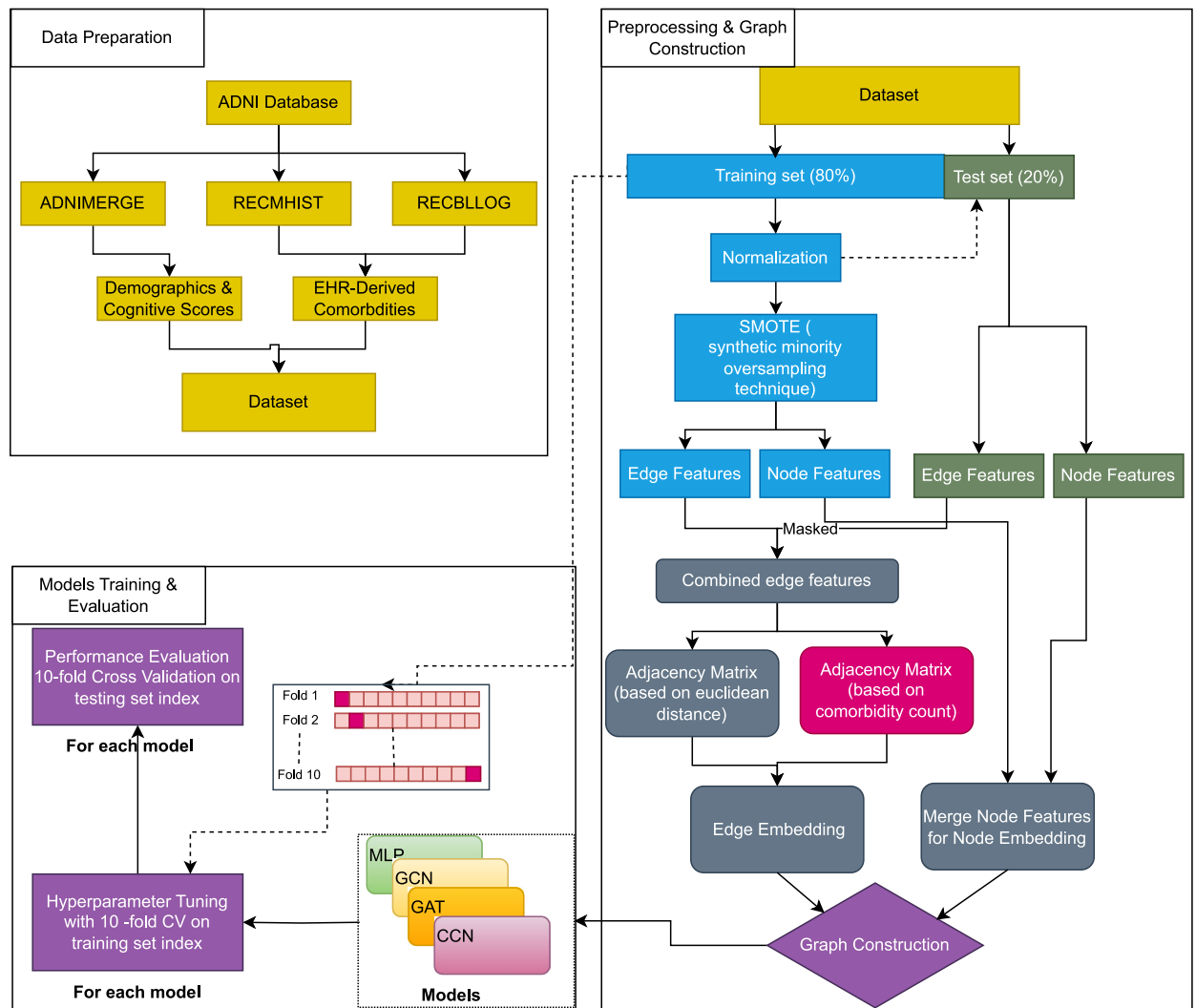


Figure 2. Overview of the pipeline of the graph neural network machine learning system.

through optimization in the range of 2-10 edge features and 60-90% similarity, as there is no systematic method providing an optimal threshold value for maximizing classification output. The optimal values were a distance of 0.1 and 7 common comorbidities. The resulting similarity matrix is symmetric with the diagonal set to 0.

The next step is to perform node and edge embedding, condensing high-dimensional feature data into low-dimensional representations. These embeddings are crucial for efficient graph-based machine learning tasks. The adjacency matrix is converted to a sparse matrix, the node features are organized into a matrix and the labels are encoded. Since we're employing PyTorch for graph-based machine learning, we convert the adjacency matrix, node features, and labels into PyTorch tensors. Subsequently, these components are encapsulated into a graph data object and employed as input for our models.

Proposed GNN models

In this study, an undirected and unweighted graph is utilized to depict the relationship between individual's overall health and Alzheimer's disease. The graph functions as a structured representation of knowledge, portraying individuals as nodes with associated features. These nodes are linked by edges, signifying relationships or their absence. GNNs excel in processing graph-structured data like medical records, distinguishing between edge presence and absence, which is crucial for understanding data relationships and offering a promising approach for AD classification. For performance analysis and comparison, we use four GNN variants: graph convolutional network (GCN), the ChebConv network (CCN), graph attention network (GAT), and GraphSAGE (SAGE). The baseline model architecture for each model started with an initial GNN layer followed by a ReLU activation function and a dropout layer. To ensure stability and mitigate the issues of over-smoothing and exploding/vanishing gradients, a PairNorm layer was added after each dropout layer⁴³. This sequence of a GNN layer, ReLU activation, dropout, and PairNorm was repeated for additional layers to produce the final node embeddings. The number of layers and specific model configurations were determined through hyperparameter optimization to achieve the best performance. To further evaluate the effectiveness of GNN models, we compared them to traditional ML methods, including SVM, RF, and XGBoost. These ML models also underwent hyperparameter optimization

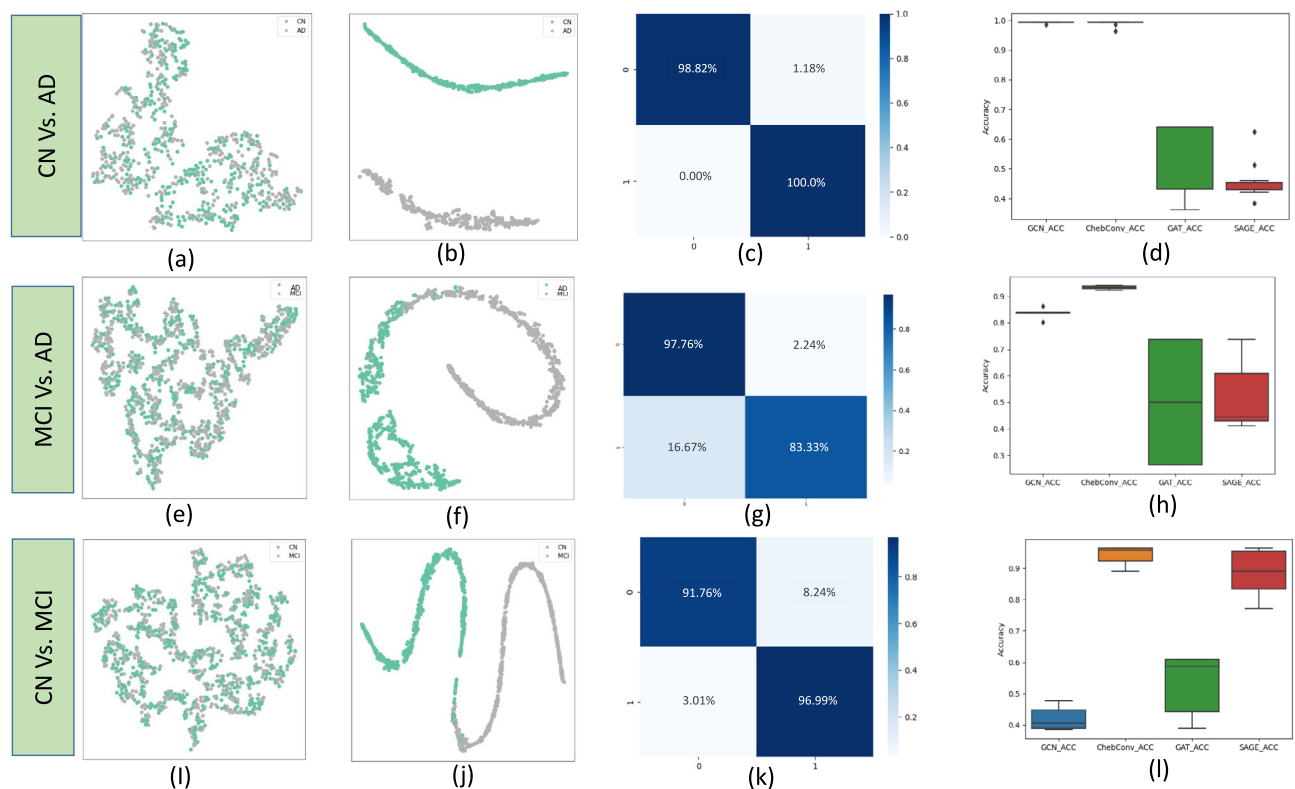


Figure 3. t-SNE visualization and performance metrics of the best performing Graph Neural Network (ChebConv) for the three binary classification tasks: CN vs. AD, MCI vs. AD, and CN vs. MCI. The first row (a–d) represents the CN vs. AD classification: (a) t-SNE visualization before training using the original feature set, (b) t-SNE visualization after training using the original feature set, (c) confusion matrix, and (d) ACC plot. The second row (e–h) represents the MCI vs. AD classification: (e) t-SNE visualization before training using the original feature set, (f) t-SNE visualization after training using the original feature set, (g) confusion matrix, and (h) ACC plot. The third row (i–l) represents the CN vs. MCI classification: (i) t-SNE visualization before training using the ablation 2 set, (j) t-SNE visualization after training using the ablation 2 set, (k) confusion matrix, and (l) ACC plot.

to ensure a fair comparison. This comparative analysis aimed to demonstrate the advantages of GNN models in handling graph-structured data for Alzheimer’s disease classification.

Graph convolution network (GCN)

Similar to the Convolution network that is used on fixed structured data such as digital images, GCN is a generalized form that could process graphs that are not fixed in structure and size. GCN uses trainable weights to learn features between nodes and their neighbors and transform the current node state (x) into a final node state (x_n) called “node embedding”. The GCN layer is defined as⁴⁴:

$$x_v^{(l+1)} = W^{(l+1)} \sum_{w \in N(v) \cup \{v\}} \frac{1}{c_{w,v}} x_w^{(l)} \quad (1)$$

where W is the trainable weight matrix, x is the node features and finally C is a fixed normalization coefficient.

ChebConv (CCN)

Another type of graph convolution networks, ChebConv⁴⁵ which employs the Chebyshev polynomials is used. Chebyshev polynomials are mathematical functions that process graph data and capture localized patterns within node neighbors in graph convolutions. In contrast to more conventional techniques based on the graph Laplacian, the ChebConv makes use of these polynomials to accomplish a type of spectral convolution that is computationally more effective. The chebyshev spectral graph convolutional operator can be represented as follows⁴⁵:

$$x' = \sum_{k=1}^K Z^k \cdot \Theta^k \quad (2)$$

Where Z^k is computed recursively by $Z^1 = X$, $Z^2 = \hat{L} \cdot X$, and $Z^k = 2 \cdot \hat{L} \cdot Z^{k-1} - Z^{k-2}$. $\hat{L}$ denotes the scaled and normalized Laplacian $\frac{2L}{L_{max}} - I$.

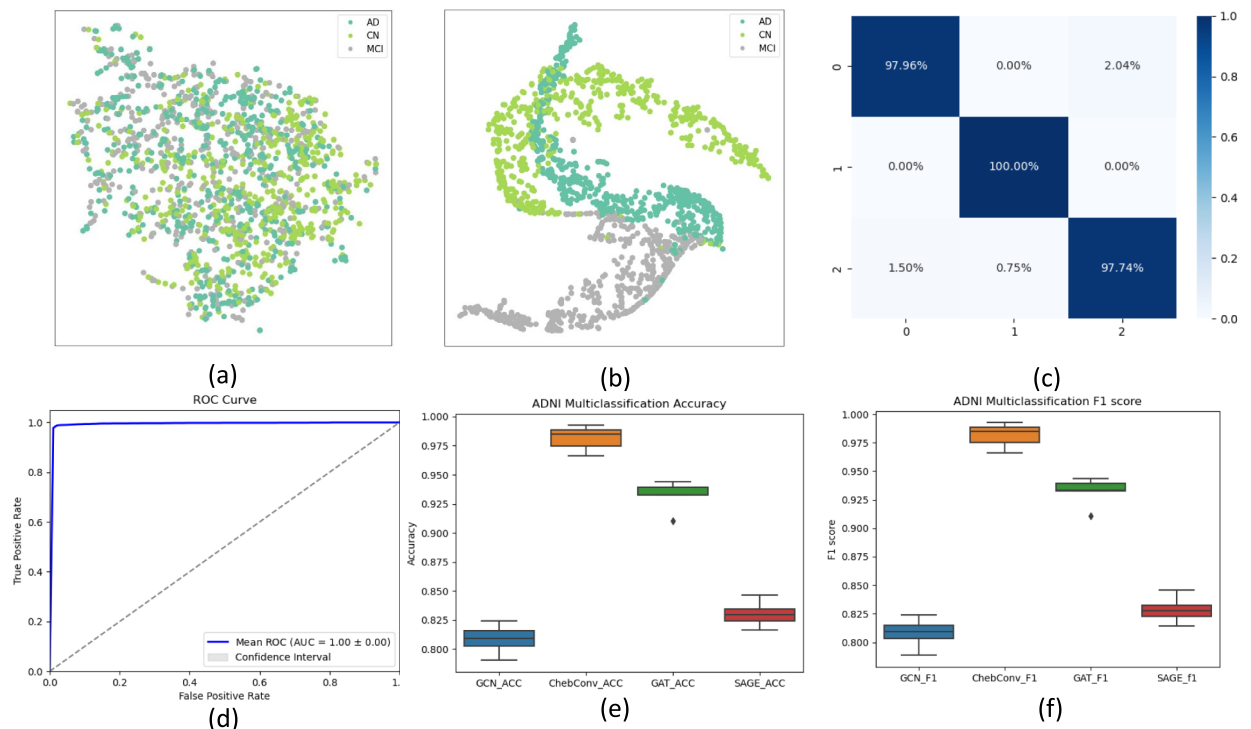


Figure 4. ChebConv Graph model performance for ADNI multi-class classification task visualized through t-SNE visualization of node representation of (a) graph inputs and (b) graph outputs. (c) Confusion matrix of multiclass classification for labels 0 (CN), 1 (MCI), and 2 (AD). The evaluation metrics with the 95% CI are seen in (d) AUC-ROC, (e) accuracy, and (f) F1-score.

Author	Algorithm	Sample no.	Data modalities	Objective	Accuracy	Strengths	Weaknesses
Li et al. 2022 ³²	Graph convolution network	168 CN 165 MCI 169 AD	Imaging (MRI, PET, DWI)	CN vs. AD CN vs. MCI MCI vs. AD CN vs. MCI vs. AD	96% 92% 95% 90%	Multi-modality	Pre-selected features
Lin et al. 2023 ³³	Convolution neural network and graph Convolution network	229 CN 382 MCI 187 AD	Imaging, demographic, and cognitive	CN vs. AD CN vs. MCI MCI vs. AD CN vs. MCI vs. AD	91.6% 96.8% 91.2% 89.4%	Handling data	Sensitive edge assignment
This work	Graph neural networks (ChebConv)	423 CN 667 MCI 241 AD	Demographic, cognitive, and EHR	CN vs. AD CN vs. MCI MCI vs. AD CN vs. MCI vs. AD	99% 94% 93% 98%	Classification accuracy & feature analysis	Sensitive feature assignment

Table 2. Performance comparison with previously published studies on using graph neural network for classification of CN, MCI and AD stages utilizing ADNI dataset.

Graph attention networks (GAT)

GATs are GNNs that introduce attention mechanisms to weigh the importance of neighboring nodes when aggregating information. The mathematical formulation of GAT is as follows⁴⁶:

$$h'_i = \sigma \left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij} W h_j \right) \quad (3)$$

$$\alpha_{ij} = \frac{\exp(\text{LeakyReLU}(\vec{a}^T [W h_i || W h_j]))}{\sum_{k \in \mathcal{N}(i)} \exp(\text{LeakyReLU}(\vec{a}^T [W h_i || W h_k]))} \quad (4)$$

Here, h_i and h_j represent the node feature vectors of nodes i and j , respectively. The attention coefficients α_{ij} are computed based on the concatenation of the transformed feature vectors $W h_i$ and $W h_j$. The attention mechanism is parameterized by learnable weights W and $\vec{a}$.

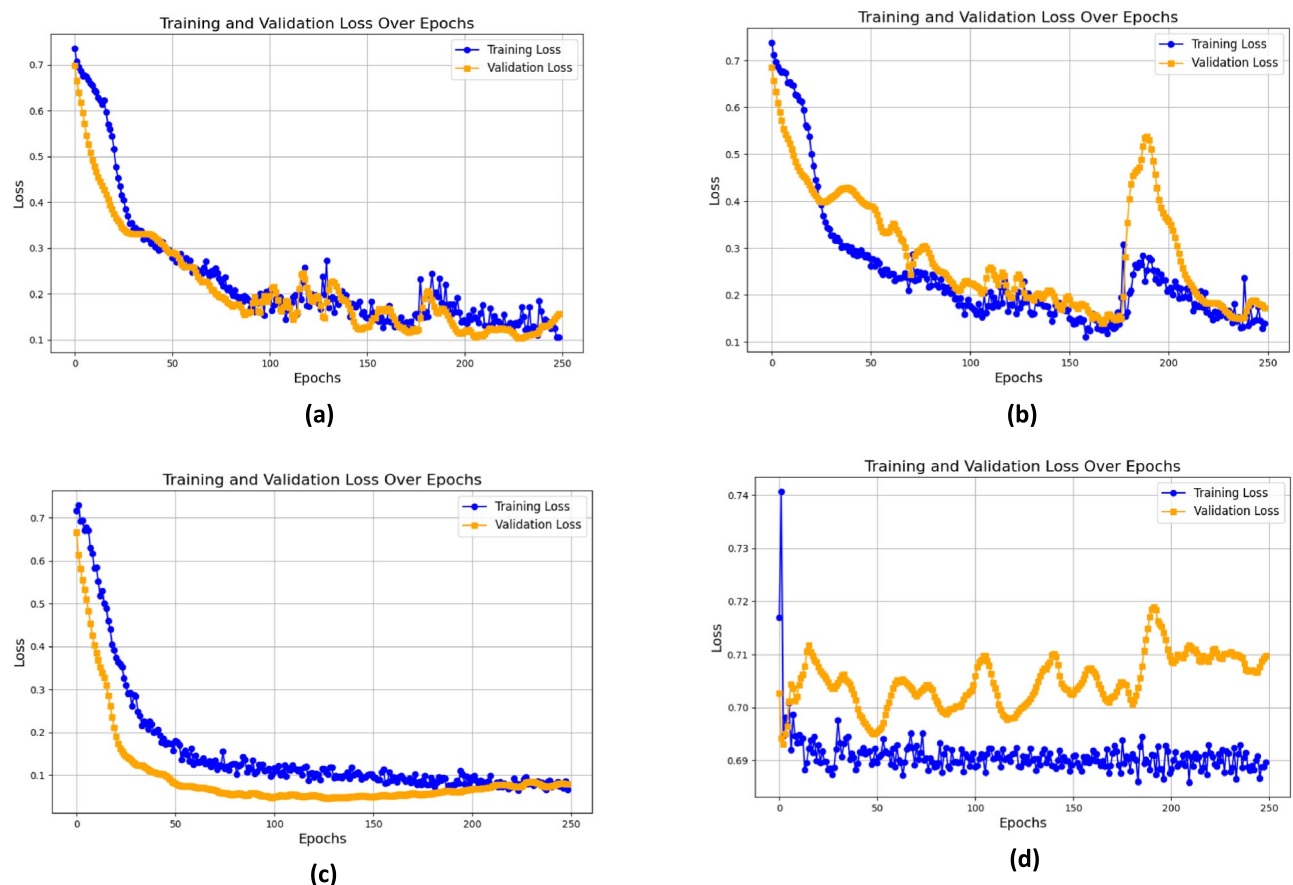


Figure 5. Loss comparison for the ChebConv model across different feature configurations in the CN vs. MCI classification task. (a) Loss with original feature set, (b) loss with Ablation Study 1 (comorbidities removed), (c) loss with Ablation Study 2 (comorbidities and cognitive scores switched), and (d) loss with Ablation Study 3 (cognitive scores removed).

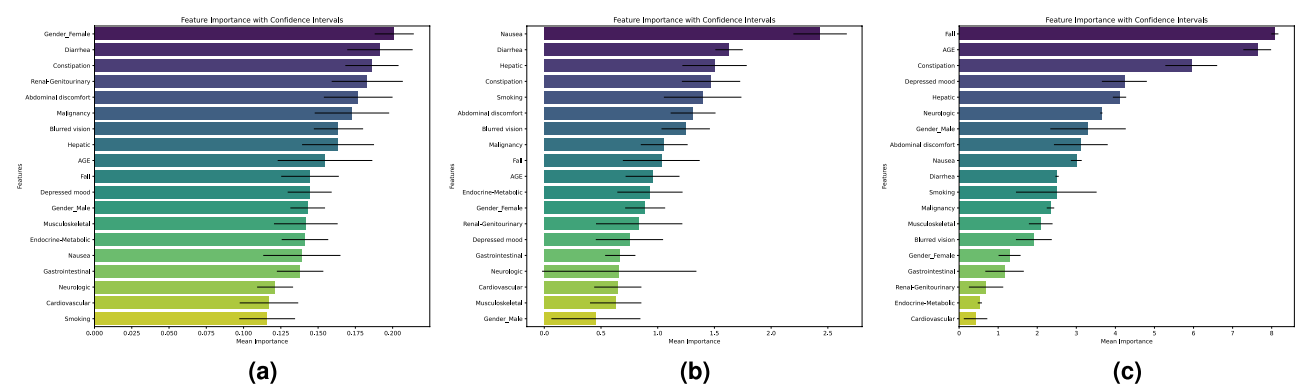


Figure 6. Bar plots of feature importance with 95% confidence interval for three binary classification tasks using the top performing ChebConv model. The plot highlights the most significant features contributing to the classification task: (a) CN vs AD, (b) MCI vs AD, and (c) for CN vs MCI. Error bars represent the 95% confidence intervals for the mean importance scores across multiple folds.

GraphSAGE

GraphSAGE (Graph Sample and Aggregate) is an inductive framework designed to generate low-dimensional embeddings for nodes in large graphs, making it highly effective for dynamic or evolving datasets⁴⁷. This method can be particularly useful for our work through enabling the integration and analysis of various health metrics, cognitive scores, and comorbidities in a scalable manner. GraphSAGE leverages a sampling and aggregation approach, where it samples a fixed-size set of neighbors for each node and aggregates their features to generate node embeddings. The aggregation function can be defined as:

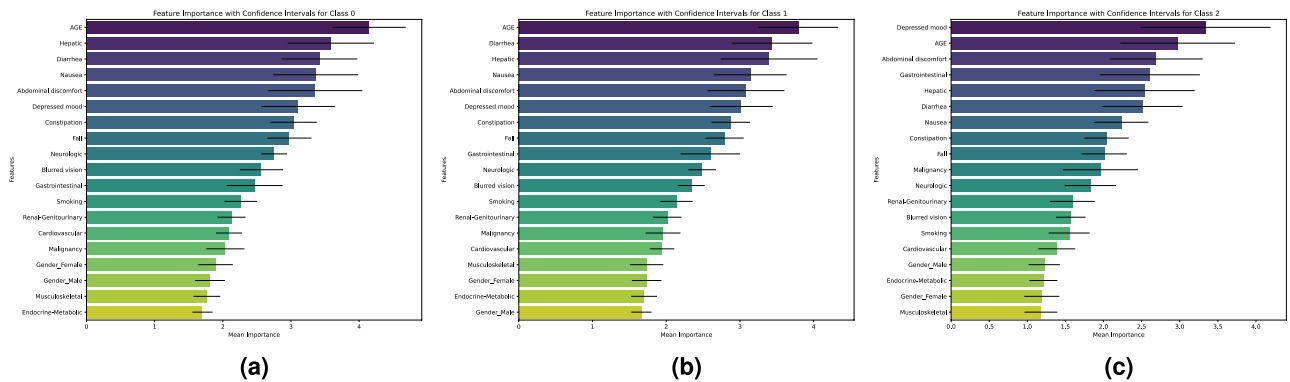


Figure 7. Bar plots display the mean importance of features by the top-performing ChebConv model on the multi classification across different classes, with error bars representing the 95% confidence intervals. The three classes are: (a) cognitive normal, (b) mild cognitive impairment, and (c) Alzheimer's disease.

GNN model	CN vs. AD		MCI vs. AD		CN vs. MCI		CN vs. MCI vs. AD	
	Study type	Accuracy	Study type	Accuracy	Study type	Accuracy	Study type	Accuracy
ChebConv	Original	0.89	Original	0.64	Ablation 2	0.84	Original	0.74

Table 3. External validation—top GNN model performance on AIBL dataset with the corresponding feature set from the top study type (original, Ablation 1, Ablation 2, or Ablation 3).

$$\mathbf{h}_v^{k+1} = \sigma(\mathbf{W} \cdot \text{AGGREGATE}(\{\mathbf{h}_u^k, \forall u \in \mathcal{N}(v)\})) \quad (5)$$

where $\mathbf{h}_v^{k+1}$ is the embedding of node v at layer $k + 1$, $\mathbf{W}$ is a weight matrix, σ is an activation function (e.g., ReLU), and $\mathcal{N}(v)$ represents the neighborhood of node v . By capturing complex relationships in health data, GraphSAGE can enhance the understanding of how various health factors contribute to the progression of Alzheimer's disease.

Hyperparameter optimization

For each model, hyperparameter optimization was performed by employing the Tree-structured Parzen Estimator (TPE) sampler algorithm⁴⁸, a Bayesian optimization technique, iteratively within a cross-validation framework consisting of 10 folds on the training set indices. TPE builds probability density functions (PDFs) to model the objective function, which in our case was minimizing the loss of the model while sampling model parameters. The PDFs are refined with every iteration to improve the accuracy of predictions. Optimal hyperparameters were determined for each model and subsequently stored for utilization in the performance evaluation. The search space for the optimized parameters in the GNN models are detailed in Table 4 and include the learning rate, dropout rate, the number of nodes in a layer, and the number of layers. The search space for the optimized parameters in the ML models are also detailed in Table 4. For RF, the search space included the number of estimators, maximum depth, minimum samples split, minimum samples leaf, and maximum features. SVM optimization involved tuning the penalty parameter (C), kernel type, and gamma value for non-linear kernels. XGBoost's search space consisted of the learning rate, maximum depth, subsample ratio, colsample by tree ratio, and the number of estimators. For convergence, a fixed number of epochs was used during the training process. Adam optimization⁴⁹ and cross-entropy loss were consistently applied across all models.

GNN implementation

This work was implemented using the R Language 4.2.1⁵⁰ with the packages tidyverse 2.0.0 and tidyr 1.3.0 for pre-processing, and Python packages such as PyTorch-Geometric (PyG) 1.13.0⁵¹ and Scikit-learn 1.3.0⁵². The detailed implementation process, including data preparation, model training, and evaluation, follows the structured approach outlined in the pseudo-code provided in Algorithm 1. The overall time complexity of the algorithm is approximately $O(k \cdot E \cdot L \cdot n^2 \cdot m^2)$, where k is the number of cross-validation folds (10), E is the number of training epochs (250), L is the number of layers in the model (hyperparameter: 1-6), n is the number of nodes (patients), and m is the number of features per node (4). The time complexity will be different for each model depending on the optimized parameter. In our work, the graph structure has defined nodes, node features, and edge features. Nodes represent individuals with the node features being their demographics and EHR-derived comorbidity data (2 demographic features and 16 comorbidity features). The edges in the graph are based on adjacency matrix that denote individuals' similarity derived from clinical tests (4 cognitive features). As mentioned in the data preparation section of this paper, the adjacency matrix has values of (0) and (1) between two nodes if the Euclidean distance between the edge feature vectors (cognitive scores) is less than 0.1. A Pytorch

Models	Parameters	Search space
GCN	Learning rate	Loguniform [0.1, 0.00001]
	Dropout rate	Uniform (0.1, 0.7)
	Number of nodes in a layer	(4, 18)
	Number of layers	(1, 6)
ChebConv	Learning rate	Loguniform [0.1, 0.00001]
	Dropout rate	Uniform (0.1, 0.7)
	Number of nodes in a layer	(4, 18)
	Number of layers	(1, 6)
	K (order of the Chebyshev polynomial)	(1, 5)
GAT	Learning rate	Loguniform [0.1, 0.00001]
	Dropout rate	Uniform (0.1, 0.7)
	Number of nodes in a layer	(4, 18)
	Number of layers	(1, 6)
	K (number of attention heads)	(1, 5)
SAGE	Learning rate	Loguniform [0.1, 0.00001]
	Dropout rate	Uniform (0.1, 0.7)
	Number of nodes in a layer	(4, 18)
	Number of layers	(1, 6)
RF	Number of estimators	(5, 50)
	Maximum depth	(1, 20)
	Minimum samples split	(2, 10)
	Minimum samples leaf	(1, 10)
	Maximum features	Categorical ['sqrt', 'log2']
SVM	Penalty parameter (C)	Loguniform [0.001, 10]
	Kernel type	Categorical ['linear', 'rbf', 'sigmoid']
	Gamma	Loguniform [0.001, 10] for 'rbf' and 'sigmoid'; 'scale' for 'linear'
XGBoost	learning rate	loguniform [0.0001, 0.3]
	Maximum depth	(3, 10)
	Subsample ratio	Uniform (0.5, 1.0)
	Colsample by tree ratio	Uniform (0.5, 1.0)
	Number of estimators	(50, 200)

Table 4. Search space of the hyper-parameters explored for each model.

graph data object with embedded nodes and edges is used as input to the graph models. The same node feature embedding is the input to the ML models.

We design our models to perform two classification tasks using the learned embeddings: multi-class classification (CN, MCI, or AD), and binary classification (CN vs. AD, CN vs. MCI, AD vs. MCI). Performing multi-class classification is important in understanding the progression and stages of cognitive decline, while binary classification enables us to discern distinctive patterns between specific disease states, providing deeper insights into cognitive impairment progression. Each model is built with the optimized parameters; in the case of the GCN multiclass classification model, it consists of two graph convolutional layers (GCNConv), each with 18 hidden units. A dropout layer with a dropout probability of approximately 11% is applied after each graph convolutional layer to prevent overfitting. ReLU activation functions are used after each dropout layer to introduce non-linearity. The output layer consists of a single graph convolutional layer with the number of target classes determined dynamically from the data. All models follow a similar architecture with their own parameters, as this architecture allows for effective learning of node representations from the graph data, with flexibility in the number of layers and hidden units, and regularization to prevent overfitting.

Study Type	Model	CN vs. AD			MCI vs. AD			CN vs. MCI			CN vs. MCI vs. AD		
		AUC	ACC	F1	AUC	ACC	F1	AUC	ACC	F1	AUC	ACC	F1
Original	SVM	0.57	0.59	0.55	0.53	0.63	0.63	0.48	0.55	0.53	0.62	0.44	0.43
	RF	0.56	0.57	0.57	0.56	0.59	0.60	0.61	0.57	0.57	0.59	0.40	0.40
	XGBoost	0.52	0.55	0.55	0.55	0.59	0.61	0.64	0.60	0.60	0.57	0.37	0.38
	GCN	0.99	0.99	0.99	0.95	0.84	0.84	0.75	0.65	0.64	0.92	0.81	0.81
	ChebConv	0.99 [0.99,0.99]	0.99 [0.98,1.0]	0.99 [0.99,0.99]	0.96 [0.96,0.97]	0.93 [0.93,0.94]	0.93 [0.93,0.94]	0.83 [0.73,0.92]	0.78 [0.67,0.89]	0.76 [0.62,0.91]	1.0 [0.99,1.0]	0.98 [0.98,0.99]	0.98 [0.98,0.99]
	GAT	0.49	0.56	0.41	0.52	0.50	0.37	0.47	0.54	0.39	0.99	0.93	0.93
	SAGE	0.51	0.45	0.43	0.57	0.51	0.50	0.5	0.52	0.47	0.94	0.83	0.83
Ablation 1	SVM	0.62	0.57	0.58	0.57	0.56	0.59	0.62	0.56	0.57	0.54	0.36	0.38
	RF	0.61	0.58	0.58	0.52	0.52	0.54	0.59	0.55	0.55	0.53	0.35	0.36
	XGBoost	0.57	0.55	0.55	0.53	0.53	0.56	0.59	0.54	0.55	0.50	0.32	0.33
	GCN	0.99	0.99	0.99	0.95	0.87	0.87	0.75	0.68	0.67	0.75	0.51	0.48
	ChebConv	0.99 [0.99,0.99]	0.99 [0.99,0.99]	0.99 [0.99,0.99]	0.96 [0.96,0.96]	0.92 [0.92,0.93]	0.92 [0.92,0.93]	0.90 [0.86,0.94]	0.85 [0.83,0.88]	0.85 [0.83,0.88]	0.99 [0.99,0.99]	0.93 [0.92,0.93]	0.93 [0.92,0.93]
	GAT	0.53	0.58	0.44	0.49	0.50	0.37	0.52	0.48	0.32	0.98	0.91	0.91
	SAGE	0.53	0.50	0.40	0.53	0.53	0.45	0.55	0.51	0.39	0.86	0.64	0.63
Ablation 2	SVM	0.99	0.99	0.99	0.93	0.93	0.93	0.97	0.93	0.93	0.97	0.91	0.91
	RF	0.99	0.99	0.99	0.95	0.92	0.92	0.97	0.94	0.94	0.99	0.94	0.94
	XGBoost	0.99	0.99	0.99	0.95	0.91	0.91	0.97	0.94	0.94	0.99	0.93	0.93
	GCN	0.47	0.46	0.45	0.54	0.31	0.21 0.50	0.42	0.34	0.84	0.63	0.55	0.54
	ChebConv	0.99 [0.99,1.0]	0.99 [0.98,0.99]	0.99 [0.98,0.99]	0.98 [0.98,0.98]	0.93 [0.93,0.94]	0.93 [0.93,0.94]	0.99 [0.99,0.99]	0.94 [0.92,0.96]	0.94 [0.92,0.96]	0.99 [0.99,1.0]	0.95 [0.94,0.96]	0.95 [0.94,0.96]
	GAT	0.64	0.59	0.51	0.53	0.60	0.52	0.51	0.53	0.44	0.99	0.94	0.94
	SAGE	0.99	0.98	0.98	0.97	0.87	0.87	0.99	0.88	0.89	0.83	0.65	0.59
Ablation 3	SVM	0.62	0.57	0.58	0.57	0.56	0.59	0.62 [0.61, 0.63]	0.56 [0.55, 0.58]	0.57 [0.55, 0.58]	0.54	0.36	0.38
	RF	0.61 [0.6, 0.62]	0.58 [0.56, 0.59]	0.58 [0.57, 0.6]	0.52	0.52	0.54	0.59	0.55	0.55	0.53	0.34	0.36
	XGBoost	0.57	0.55	0.55	0.53	0.53	0.56	0.59	0.54	0.55	0.50	0.32	0.33
	GCN	0.47	0.46	0.44	0.54	0.29	0.16	0.50	0.45	0.35	0.57	0.45	0.38
	ChebConv	0.55	0.56	0.56	0.61	0.46	0.46	0.60	0.53	0.51	0.60	0.42	0.42
	GAT	0.49	0.52	0.42	0.53 [0.52,0.54]	0.59 [0.44,0.74]	0.51 [0.34,0.68]	0.51	0.52	0.39	0.59	0.40	0.40
	SAGE	0.47	0.49	0.40	0.53	0.44	0.39	0.59	0.52	0.44	0.65 [0.63,0.66]	0.47 [0.43,0.51]	0.34 [0.31,0.38]

Table 5. Model performance comparison for ADNI data, showcasing the accuracy (ACC), area under the curve (AUC), and F1-score (F1) for each classification task. Bold values indicate the best performance for that feature set, with 95% confidence intervals provided only for these results. *RF* random forest, *SVM* support vector machine, *GCN* graph convolutional network, *ChebConv* Chebyshev convolution, *GAT* graph attention network, *SAGE* GraphSAGE, *XGBoost* extreme gradient boosting.

- 1: **Load** ADNI and AIBL
- 2: **Groupby** patient ID and perform one-hot encoding categorical features
- 3: **train_test_split** stratified splitting
- 4: **MinMaxScaler** Normalize training and carry to test sets then apply **SMOTE**.
- 5: Define and separate **node** and **edge** features based on StudyType
- 6: **Construct** adjacency matrix based for edges and **convert** matrices to sparse format
- 7: Embed graph using node features and adjacency matrix
- 8: Define model architecture, and **train** using CV
- 9: Evaluate model on test sets (accuracy, F1-score, ROC, AUC)

Algorithm 1. Pseudo-code for GNN implementation.

Evaluation metrics
To evaluate the classification performance of our GNN models, we utilized multiple metrics, including area under the receiver operating curve (AUC-ROC), accuracy (ACC), and F1 score (F1). Employing these metrics

allows for a thorough assessment of the models' effectiveness in distinguishing between different cognitive states. Before cross-validation, GNN models were tuned to optimized parameters for peak performance. A rigorous tenfold cross-validation on testing set indices was then conducted to enhance reliability and minimize bias in our evaluations.

$$\text{AUC-ROC} = \int_0^1 \text{TPR}(\text{FPR}^{-1}(x)) dx \quad (6)$$

$$\text{Accuracy (ACC)} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (7)$$

$$\text{F1 Score (F1)} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (8)$$

The evaluation was conducted using a tenfold cross-validation approach on the testing set indices. In addition to cross-validation, external validation was performed using the AIBL dataset. This external validation further tested the models' performance, providing insights into their generalizability across different cohorts.

Model interpretation

GNNs are gaining popularity for processing graph-structured data. However, the complexity of graph-based data and the non-linear relationships between nodes can obscure the rationale behind GNN predictions. Consequently, there is a growing interest in understanding their prediction mechanisms. In our models, we are incorporating different sets of node and edge features. Calculating importance scores based on node features allows us to understand the influence of individual attributes on the classification of CN, MCI, and AD classes. This approach indicates the significance of individual node features in making predictions for each class. On the other hand, importance scores derived from edge features highlights the relevance of connections or relationships between nodes in predicting each class. Our primary interest lies in understanding the importance of individual features for classification purposes. Therefore, selecting node features aligns with the specific objectives of our study and the desired depth of analysis into the influence of each feature on class predictions.

The calculation of importance scores for The importance scores were calculated as the mean of the gradient-based sensitivity of the model's output with respect to each input feature, averaged across 10 cross-validation folds. This approach provides a robust measure of feature importance, capturing the consistency of each feature's influence across different subsets of the data. Furthermore, confidence intervals (CI) were computed for these mean importance scores, offering additional insight into the reliability and variability of the importance estimates. The calculation process involved computing gradients during the forward pass, where input features undergo transformations through multiple layers, resulting in output logits for each class. During backpropagation, the gradients of the output logits with respect to the input features were calculated as follows:

$$\nabla_{\mathbf{x}} \text{Loss} = \frac{\partial \text{Loss}}{\partial \text{Logits}} \cdot \frac{\partial \text{Logits}}{\partial \mathbf{x}} \quad (9)$$

Here, $\nabla_{\mathbf{x}} \text{Loss}$ represents the gradient of the loss function with respect to the input features ($\mathbf{x}$), which we seek to compute. The gradient is obtained by multiplying the gradient of the loss with respect to the logits ($\frac{\partial \text{Loss}}{\partial \text{Logits}}$) and the gradient of the logits with respect to the input features ($\frac{\partial \text{Logits}}{\partial \mathbf{x}}$).

This gradient-based approach, also known as a saliency map, allows us to assess how variations in input features influence the model's predictions⁵³. The resulting absolute gradient values represent the magnitude of this sensitivity, with higher values indicating a greater influence on the model's decisions. By aggregating these values across all folds, we derived the mean importance scores for each feature, alongside their corresponding confidence intervals, providing a comprehensive understanding of feature importance and its stability in the classification process.

Ablation studies

Ablation studies in GNNs involve systematically altering components such as node and edge features to discern their individual impact on the model's performance. This is achieved by interchanging the roles of edge and node features, and subsequently, by modifying the set of features utilized in the nodes. These ablations serve critical purposes: they shed light on the significance of distinct feature sets. By systematically testing these alterations, we can pinpoint the most influential components, thereby identifying what could be driving the model's performance and informing future improvements to the architecture. We conducted three ablation studies to investigate the impact of altering network features. Initially, the original setup comprised node features representing demographics and comorbidities, with cognitive scores as edge features. In ablation 1, we removed comorbidity features from the node feature vector. In ablation 2, we interchanged the features, with node features representing demographics and cognitive scores, and edge features representing comorbidities. This design is particularly intriguing as it explores whether comorbidities better connect patients compared to cognitive scores. Finally, in ablation 3, we maintained the previous network design but excluded cognitive scores from the node features.

Conclusion

In conclusion, this paper introduces a novel GNN approach tailored for multi-class and binary classification tasks involving AD, CN, and MCI based on clinical data. Our findings demonstrate that incorporating comorbidity data into the feature set yields performance levels comparable to leading models, even outperforming those utilizing MRI data alone and traditional machine learning methods such as SVM, RF, and XGBoost. Moreover, the GNN models performed well when validated on an external test set, AIBL, showing their generalizability. The integration of a comprehensive feature set emerges as pivotal, showcasing superior overall performance, consistent with prior studies emphasizing the significance of longitudinal clinical EHR data in monitoring AD progression. In a complex disease like AD, considering comorbidities becomes essential as they have been linked to AD through underlying pathways, such as inflammation. This highlights the importance of incorporating a broad range of clinical data, including comorbidities, to gain deeper insights into disease progression and its associated factors. Moving forward, our future endeavors will focus on further refining our approach by integrating longitudinal clinical data into GNN models and exploring alternative architectural designs to enhance classification accuracy and robustness.

Data availability

The dataset analyzed in this study can be accessed from the Alzheimer's Disease Neuroimaging Initiative (ADNI) repository at adni.loni.usc.edu (Accession Number: sa000002). Data from the Australian Imaging Biomarkers and Lifestyle flagship study of ageing (AIBL) are also available at the ADNI database (www.aibl.csiro.au/adni/index.html). Access to both datasets requires a simple application procedure that is readily granted to researchers.

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Author contributions

AMS acquired funding and supervised the project. FA and MKA conceptualized the study, performed the literature review, built the models, analyzed the data, generated visualizations, and wrote the initial manuscript. MKA refined the models and conducted feature analysis. FA provided the interpretation and wrote the discussion. AMS offered expert opinions and verified the methods. All authors reviewed the manuscript.

Competing Interests

The authors declare no competing interests.

Additional information

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